

TRANSERIES / SINGULAR INVERSION / COUNTABLE EXPONENTIAL FEEDBACK

# Confluent Critical Transseries

## Across the Exponent-Two Boundary

A Uniform Inverse Chart, Conditional Poisson Limits,  
and Sharp Action Budgets

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### Abstract

We study the positive feedback equation

$$U_\varepsilon(q) = c_\varepsilon \sum_{j \geq 1} j^{-3+\varepsilon} q^j e^{jU_\varepsilon(q)}, \quad c_\varepsilon = \zeta(2-\varepsilon)^{-1},$$

in the joint limit in which the coefficient order tends to infinity and  $\varepsilon \rightarrow 0$ , from either side and at an arbitrary rate. A cancellation-free normal form yields a single convergent three-variable inverse chart over an exact implicit core. At  $\varepsilon = 0$  this core is a Lambert- $W_{-1}$  core; its two outer limits recover fractional-power and finite-variance inversion. We prove a triangular local Gaussian theorem, stable under arbitrary thinning above the extreme-action scale. It gives a multiplicative equivalent for the sector coefficients and a conditional Poisson point-process limit even under exact lattice conditioning. The fluctuation scale  $B_n$  and extreme scale  $A_n$  satisfy  $A_n/B_n \rightarrow 0$  for every  $\varepsilon_n \rightarrow 0$ , while the sharp retained-coefficient fraction at cutoff  $M \sim sA_n$  is  $\exp(-1/(2s^2))$ . Explicit crossover factors describe the window  $\varepsilon_n \log n = O(1)$ . An exact identity between the inverse core and the probabilistic normalization connects the two constructions. Complete proofs, finite real inequalities, exact algebraic checks, and reproducible numerical diagnostics are included.

**Research status and boundary.** This article addresses the *upper-endpoint joint-limit part* of Research Question 4 in the repository’s countable-action critical-Hahn article. The fixed logarithmic endpoint was already treated in subsequent drafts. The present contribution is the uniform confluent construction and its arbitrary-rate, locally conditioned limit theory for the specified model. The classical Gaussian/extreme-value separation at a fixed cubic tail is not claimed as new. Publication priority has not been established by an exhaustive literature review. No Lean formalization, outward-rounded implementation, or resolution of unrestricted transseries realization is claimed.

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# 1 The research target and the main conclusions

## 1.1 A precise continuation of the repository program

The inspected `ProveIt` research index separates its formal Lean library from a collection of recent, unmerged research packages [1]. The article *Beyond Finite-Action Folds* treats positive action weights with exponent  $1 + \alpha$ , for fixed  $1 < \alpha < 2$ . Its Research Question 4 asks for joint limits in which  $\alpha$  approaches an endpoint while the sector order grows, specifically warning against substituting into formulas containing  $\Gamma(-\alpha)$  [2].

Two later drafts treat the fixed upper endpoint, where the tail is exactly inverse cubic outside a finite prefix [3, 4]. The first explicitly excludes a simultaneous moving-exponent limit. We set  $\alpha = 2 - \varepsilon$ , allow either sign of  $\varepsilon$ , and solve a normalized, exact-tail version of that remaining problem. The restriction to an exact power tail makes it possible to combine an explicit analytic chart with a self-contained Fourier proof. A finite-prefix robustness result is proved separately; no general slowly varying triangular array is silently included.

The existing manuscripts are used to identify a research target, not as unverified premises. The coefficient representation, normal form, scale estimates, local limits, and cutoff conclusions below are proved afresh. The inspected repository revision and the limits of this comparison are recorded in Appendix A.

## 1.2 What is proved

There are two natural problems. The first is local inversion near the critical value of  $q$ ; the second is the asymptotic size of  $[q^n]U_\varepsilon$ . They are controlled by the same elementary function

$$h_\varepsilon(x) = \int_1^x y^{-1+\varepsilon} dy = \begin{cases} (x^\varepsilon - 1)/\varepsilon, & \varepsilon \neq 0, \\ \log x, & \varepsilon = 0. \end{cases} \quad (1.1)$$

The analytic result is an *exact convergent chart*, not merely a finite asymptotic approximation. For the small positive root of  $\Psi_\varepsilon(t) = \delta$ , it has the form

$$t = r \mathcal{V}\left(r, \varepsilon, \frac{1}{h_\varepsilon(1/r)}\right), \quad \frac{c_\varepsilon}{2} r^2 h_\varepsilon(1/r) = \delta, \quad (1.2)$$

where one holomorphic function  $\mathcal{V}$  works near the origin in three independent variables. Its coefficients have a finite recursive construction and a geometric tail bound.

For any sequence  $\varepsilon_n \rightarrow 0$ , the probabilistic scales are

$$A_n = (nc_{\varepsilon_n})^{1/(2-\varepsilon_n)}, \quad B_n^2 = nc_{\varepsilon_n} h_{\varepsilon_n}(B_n), \quad (1.3)$$

with the large positive solution chosen for  $B_n$ . We prove  $A_n/B_n \rightarrow 0$ , a uniform lattice local limit on scale  $B_n$ , and a conditional Poisson limit on scale  $A_n$ . Consequently,

$$[q^n]U_{\varepsilon_n}(q) \sim \frac{\rho_{\varepsilon_n}^{-n}}{\sqrt{2\pi} n B_n}, \quad \frac{[q^n]U_{\varepsilon_n}^{[M_n]}(q)}{[q^n]U_{\varepsilon_n}(q)} \rightarrow e^{-1/(2s^2)} \quad \text{if } M_n/A_n \rightarrow s > 0. \quad (1.4)$$

The exponential factor  $\rho_{\varepsilon_n}^{-n}$  must be kept at the actual parameter value. Substituting  $\rho_0$  there is generally invalid even when  $\varepsilon_n \rightarrow 0$ .

## 1.3 Classical background and novelty qualifications

Lagrange inversion, Poisson coefficient representations, the analytic implicit function theorem, and Fourier inversion are standard tools. Janson's Example 18.29 gives a classical fixed cubic-tail

instance of Gaussian fluctuations decoupling from extremes under conditioning [6]. Our point is not to rediscover that endpoint mechanism. It is to prove a single result for a moving exponent, with no restriction on the rate or sign of its approach, and to connect it to a convergent singular-inversion chart. We supply the triangular local-limit argument instead of assuming that a fixed-distribution theorem is uniform in a changing exponent.

The formulas constitute model-specific theorems with proofs. Whether their precise combination, or any component in an equivalent normalization, has appeared elsewhere remains a bibliographic question. In particular, “confluent” here describes the merging of the fractional-power and logarithmic descriptions; it is not a new axiom system for transseries.

## 2 The normalized model and exact coefficient identities

Throughout the real asymptotic arguments,  $|\varepsilon| < 1/4$ ,  $\alpha = 2 - \varepsilon$ , and

$$c_\varepsilon = \frac{1}{\zeta(\alpha)}, \quad F_\varepsilon(z) = c_\varepsilon \operatorname{Li}_{1+\alpha}(z) = \sum_{j \geq 1} a_{\varepsilon,j} z^j, \quad a_{\varepsilon,j} = c_\varepsilon j^{-1-\alpha}. \quad (2.1)$$

Thus  $\sum j a_{\varepsilon,j} = 1$ . It is useful to include a coupling parameter:

$$U_{\varepsilon,\beta}(q) = \beta F_\varepsilon(q e^{U_{\varepsilon,\beta}(q)}), \quad \beta > 0. \quad (2.2)$$

The notation without  $\beta$  means  $\beta = 1$ . A cutoff superscript  $[M]$  means that only actions  $1 \leq j \leq M$  are retained. All formal solutions have zero constant coefficient. Successive coefficient extraction in (2.2) proves their uniqueness.

**Proposition 2.1** (Lagrange and Poisson identities). *For  $n \geq 1$ , let  $u_n(\varepsilon, \beta) = [q^n]U_{\varepsilon,\beta}(q)$ . Then*

$$u_n(\varepsilon, \beta) = \frac{1}{n} [z^n] e^{n\beta F_\varepsilon(z)}. \quad (2.3)$$

*Let  $Z_{n,j}$  be independent Poisson variables of means  $\nu_{n,j} = n\beta a_{\varepsilon,j}$ , and put  $S_n = \sum j Z_{n,j}$ . This sum is finite almost surely, and*

$$u_n(\varepsilon, \beta) = \frac{e^{n\beta F_\varepsilon(1)}}{n} \Pr(S_n = n). \quad (2.4)$$

*If  $S_{n,M} = \sum_{j \leq M} j Z_{n,j}$  and  $\Lambda_{n,M} = \sum_{j > M} \nu_{n,j}$ , then*

$$R_n(M) := \frac{u_n^{[M]}(\varepsilon, \beta)}{u_n(\varepsilon, \beta)} = e^{-\Lambda_{n,M}} \frac{\Pr(S_{n,M} = n)}{\Pr(S_n = n)} \quad (2.5)$$

$$= \Pr(Z_{n,j} = 0 \text{ for all } j > M \mid S_n = n). \quad (2.6)$$

*In particular,  $R_n(M)$  is nondecreasing, belongs to  $(0, 1]$ , and equals one for  $M \geq n$ .*

*Proof.* Set  $z = q e^U$ . Then  $z = q e^{\beta F_\varepsilon(z)}$ , and the usual formal Lagrange formula gives

$$[q^n] \beta F_\varepsilon(z(q)) = \frac{\beta}{n} [z^{n-1}] F'_\varepsilon(z) e^{n\beta F_\varepsilon(z)}.$$

Differentiating the exponential proves (2.3). The total Poisson intensity is finite, so only finitely many Poisson events occur almost surely. The probability generating function of their weighted sum is  $\exp(n\beta(F_\varepsilon(z) - F_\varepsilon(1)))$ , proving (2.4). Independence of retained and deleted actions gives the two ratio identities. The positivity conclusions also follow directly from (2.3); actions exceeding  $n$  cannot affect its  $n$ -th coefficient.  $\square$

At critical coupling define

$$U_{\varepsilon,c} = F_\varepsilon(1), \quad \rho_\varepsilon = e^{-F_\varepsilon(1)}. \quad (2.7)$$

For  $z = e^{-t}$ ,  $q = \rho_\varepsilon e^{-\delta}$ , the exact boundary equation is

$$\delta = \Psi_\varepsilon(t) := t + F_\varepsilon(e^{-t}) - F_\varepsilon(1) \quad (2.8)$$

$$= c_\varepsilon \sum_{j \geq 1} j^{-1-\alpha} (e^{-jt} - 1 + jt),$$

$$U_{\varepsilon,c} - U_\varepsilon(q) = t - \delta. \quad (2.9)$$

For  $t > 0$ , termwise differentiation gives

$$\Psi'_\varepsilon(t) = c_\varepsilon \sum_{j \geq 1} j^{-\alpha} (1 - e^{-jt}) > 0, \quad \Psi''_\varepsilon(t) = c_\varepsilon \sum_{j \geq 1} j^{1-\alpha} e^{-jt} > 0. \quad (2.10)$$

Also  $\Psi'_\varepsilon(t) \rightarrow 0$  as  $t \downarrow 0$ , by dominated convergence using  $\sum j^{-\alpha} < \infty$ .

**Proposition 2.2** (The critical radius). *The radius of convergence of  $U_\varepsilon$  is exactly  $\rho_\varepsilon$ , and its real boundary value is  $U_{\varepsilon,c}$ .*

*Proof.* The map  $q(z) = ze^{-F_\varepsilon(z)}$  has positive derivative on  $(0, 1)$ , since  $zF'_\varepsilon(z) < F'_\varepsilon(1) = 1$ . It maps that interval onto  $(0, \rho_\varepsilon)$ , giving the claimed branch and boundary value. Identity (2.4) bounds  $u_n(\varepsilon) \leq \rho_\varepsilon^{-n}/n$ , so the radius is at least  $\rho_\varepsilon$ . Equations (2.9) and (2.10) show that the derivative of the inverse branch becomes unbounded at the boundary, since  $dt/d\delta = 1/\Psi'_\varepsilon(t)$ . The branch is therefore not analytic at  $\rho_\varepsilon$ , proving equality.  $\square$

### 3 A cancellation-free normal form

Define the entire crossover factor

$$\mathcal{E}(y) = \begin{cases} (e^y - 1)/y, & y \neq 0, \\ 1, & y = 0. \end{cases} \quad h_\varepsilon(x) = \log x \mathcal{E}(\varepsilon \log x). \quad (3.1)$$

For  $x > 1$ , let

$$v_\varepsilon(x) = \frac{1}{h_\varepsilon(x)}, \quad d_\varepsilon(x) = \frac{x^\varepsilon}{h_\varepsilon(x)} = \varepsilon + v_\varepsilon(x). \quad (3.2)$$

Both quantities are positive, including when  $\varepsilon < 0$ .

**Lemma 3.1** (Uniform elementary estimates). *For real  $|\varepsilon| < 1/4$  and  $x > 1$ ,*

$$v_\varepsilon(x) + d_\varepsilon(x) \leq |\varepsilon| + \frac{2}{\log x}. \quad (3.3)$$

If  $J = \lfloor x \rfloor$ , then

$$0 \leq \sum_{j=1}^J j^{-1+\varepsilon} - h_\varepsilon(x) \leq 1. \quad (3.4)$$

Uniformly as  $\varepsilon \rightarrow 0$  and  $t \downarrow 0$ ,

$$\Psi_\varepsilon(t) = \frac{c_\varepsilon}{2} t^2 h_\varepsilon(1/t) \left( 1 + O \left( |\varepsilon| + \frac{1}{\log(1/t)} \right) \right), \quad (3.5)$$

$$\Psi'_\varepsilon(t) = c_\varepsilon t h_\varepsilon(1/t) \left( 1 + O \left( |\varepsilon| + \frac{1}{\log(1/t)} \right) \right). \quad (3.6)$$

*Proof.* Write  $L = \log x$ . For  $\varepsilon \geq 0$ ,  $h_\varepsilon(x) \geq L$ , so  $v \leq 1/L$  and  $d = \varepsilon + v$ . For  $\varepsilon < 0$ , put  $a = -\varepsilon$ ; then  $d = a/(e^{aL} - 1) \leq 1/L$  and  $v = a + d$ . This proves (3.3). The function  $y^{-1+\varepsilon}$  is decreasing. Integrating on the intervals on either side of each integer gives  $\int_1^{J+1} f \leq \sum_{j \leq J} f(j) \leq 1 + \int_1^J f$ , which implies (3.4) even when  $x$  is not integral.

For the remaining estimates split each series at  $J = \lfloor 1/t \rfloor$ . On the lower part,  $e^{-u} - 1 + u = u^2/2 + O(u^3)$ , with an absolute constant for  $0 \leq u \leq 1$ . The error is  $O(t^3 \sum_{j \leq J} j^{2-\alpha}) = O(t^\alpha)$ , uniformly in the indicated exponent interval. On the upper part,  $0 \leq e^{-u} - 1 + u \leq u$ , so its contribution is also  $O(t^\alpha)$ . The quadratic lower sum is  $t^2(h_\varepsilon(1/t) + O(1))/2$ . Relative to the main term the errors are  $O(v_\varepsilon(1/t) + d_\varepsilon(1/t))$ . The derivative follows from  $1 - e^{-u} = u + O(u^2)$  below the split and  $1 - e^{-u} \leq 1$  above it, with errors  $O(t^{\alpha-1})$ . Apply (3.3).  $\square$

The elementary estimates already give the leading crossover. To obtain a convergent chart we need an exact normal form. Define functions with their removable values at zero:

$$g(\varepsilon) = \Gamma(\varepsilon - 2) - \frac{1}{2\varepsilon}, \quad b(\varepsilon) = \zeta(1 - \varepsilon) + \frac{1}{\varepsilon}, \quad (3.7)$$

$$\mathcal{R}(\varepsilon, t) = \sum_{k \geq 3} \frac{(-1)^k \zeta(3 - \varepsilon - k)}{k!} t^{k-3}. \quad (3.8)$$

The first values, with  $\gamma_1$  the first Stieltjes constant, are

$$\begin{aligned} g_0 &:= g(0) = \frac{3}{4} - \frac{\gamma}{2}, & g_1 &:= g'(0) = \frac{7}{8} - \frac{3\gamma}{4} + \frac{\gamma^2}{4} + \frac{\pi^2}{24}, \\ b(0) &= \gamma, & b'(0) &= \gamma_1, & \mathcal{R}(0, 0) &= \frac{1}{12}. \end{aligned} \quad (3.9)$$

Our sign convention is  $\zeta(1 + s) = 1/s + \sum_{k \geq 0} (-1)^k \gamma_k s^k / k!$ .

**Proposition 3.2** (Exact confluent normal form). *The functions  $g$  and  $b$  are holomorphic near zero, and  $\mathcal{R}$  is holomorphic on a product neighborhood of  $(0, 0)$ . For small positive  $t$  and real  $\varepsilon$  near zero,*

$$\frac{\Psi_\varepsilon(t)}{c_\varepsilon} = \frac{t^2}{2} [h_\varepsilon(1/t) + 2g(\varepsilon)t^{-\varepsilon} + b(\varepsilon)] + t^3 \mathcal{R}(\varepsilon, t). \quad (3.10)$$

In particular,

$$\frac{\Psi_0(t)}{c_0} = \frac{t^2}{2} \left( \log \frac{1}{t} + \frac{3}{2} \right) + \frac{t^3}{12} - \frac{t^4}{288} + O(t^6), \quad c_0 = \frac{6}{\pi^2}. \quad (3.11)$$

*Proof.* For nonzero  $\varepsilon$  near zero, the standard polylogarithm expansion [5, Eq. 25.12.12] gives

$$\text{Li}_{3-\varepsilon}(e^{-t}) = \Gamma(\varepsilon - 2)t^{2-\varepsilon} + \sum_{k \geq 0} \frac{\zeta(3 - \varepsilon - k)(-t)^k}{k!}, \quad |t| < 2\pi,$$

on the chosen logarithm branch. The constant and linear terms cancel in (2.8). Separate the two poles as in (3.7); their combined contribution is exactly  $t^2 h_\varepsilon(1/t)/2$ . This proves (3.10) away from zero. The Laurent expansions of  $\Gamma$  and  $\zeta$  give (3.9) and the removable extensions.

For completeness, the tail in (3.8) converges locally uniformly in both variables. The functional equation of  $\zeta$ , applied at  $3 - \varepsilon - k$ , bounds its coefficient divided by  $k!$  by  $Ck^C(2\pi)^{-k}$  on a sufficiently small closed complex  $\varepsilon$ -disc; the finitely many initial terms cause no difficulty. The Weierstrass theorem then gives joint holomorphy for  $|t| < 2\pi$  on smaller product domains. Continuity in  $\varepsilon$  extends (3.10) to zero. Finally,  $\zeta(0) = -1/2$ ,  $\zeta(-1) = -1/12$ , and  $\zeta(-2) = 0$  give (3.11).  $\square$

*Remark 3.3.* The separate terms  $\Gamma(\varepsilon - 2)t^{2-\varepsilon}$  and  $\zeta(1 - \varepsilon)t^2/2$  have opposite divergent poles. Retaining only the first one is not uniform when  $\varepsilon \log(1/t)$  remains bounded. The function  $h_\varepsilon$  retains their cancellation exactly, including at  $\varepsilon = 0$ .

## 4 A single convergent inverse chart

### 4.1 The exact core and its three small parameters

For  $\delta > 0$  small, let  $r = r_\varepsilon(\delta) \in (0, e^{-2})$  solve

$$\frac{c_\varepsilon}{2} r^2 h_\varepsilon(1/r) = \delta, \quad v = \frac{1}{h_\varepsilon(1/r)}. \quad (4.1)$$

Existence and uniqueness hold uniformly for  $|\varepsilon|$  sufficiently small. Indeed, the left side tends to zero at  $r = 0$  and its logarithmic derivative with respect to  $r$  is  $2 - d_\varepsilon(1/r) > 0$  on this interval, by (3.3). At the endpoint,

$$r_0(\delta) = \exp \left[ \frac{1}{2} W_{-1} \left( -\frac{4\delta}{c_0} \right) \right]. \quad (4.2)$$

The three variables  $(r, \varepsilon, v)$  tend to zero in every joint limit  $\delta \downarrow 0$ ,  $\varepsilon \rightarrow 0$ , however slowly the exponent moves.

Let

$$Q_\varepsilon(V) = \frac{V^{-\varepsilon} - 1}{\varepsilon}, \quad Q_0(V) = -\log V, \quad (4.3)$$

where  $\log V$  is the analytic branch near  $V = 1$ . This is holomorphic in  $(\varepsilon, V)$  near  $(0, 1)$ .

**Theorem 4.1** (Convergent confluent chart). *There is a holomorphic function  $\mathcal{V}(r, \varepsilon, v)$ , on one neighborhood of  $(0, 0, 0)$ , with  $\mathcal{V}(0, 0, 0) = 1$ , such that the unique small positive solution of  $\Psi_\varepsilon(t) = \delta$  is exactly*

$$t = r\mathcal{V}(r, \varepsilon, v), \quad r = r_\varepsilon(\delta), \quad v = h_\varepsilon(1/r)^{-1}. \quad (4.4)$$

The defining analytic equation for  $V = \mathcal{V}(r, \varepsilon, v)$  is

$$0 = V^2 \left[ 1 + (\varepsilon + v)Q_\varepsilon(V) + 2g(\varepsilon)(\varepsilon + v)V^{-\varepsilon} + b(\varepsilon)v + 2rvV\mathcal{R}(\varepsilon, rV) \right] - 1. \quad (4.5)$$

Its Taylor series is convergent and has a finite coefficient recurrence. For suitable positive radii  $R_1, R_2, R_3$  and  $K < \infty$ , its truncation  $\mathcal{V}_{\leq N}$  by total degree satisfies

$$|\mathcal{V} - \mathcal{V}_{\leq N}| \leq K \frac{(N+3)^2 \theta^{N+1}}{(1-\theta)^3} \quad \text{when} \quad |r| \leq \theta R_1, \quad |\varepsilon| \leq \theta R_2, \quad |v| \leq \theta R_3, \quad (4.6)$$

for  $0 < \theta < 1$ .

*Proof.* Substitute  $t = rV$  in (3.10) and divide by  $\delta = c_\varepsilon r^2 h_\varepsilon(1/r)/2$ . The exact identity

$$\frac{h_\varepsilon(1/(rV))}{h_\varepsilon(1/r)} = 1 + (\varepsilon + v)Q_\varepsilon(V), \quad \frac{r^{-\varepsilon}}{h_\varepsilon(1/r)} = \varepsilon + v$$

gives (4.5). Its left side, denoted by  $\mathcal{F}(r, \varepsilon, v; V)$ , is holomorphic in a neighborhood of  $(0, 0, 0; 1)$ . At  $(r, \varepsilon, v) = (0, 0, 0)$  it is  $V^2 - 1$ , with derivative 2 at  $V = 1$ . The analytic implicit function theorem proves the existence and uniqueness of  $\mathcal{V}$ . For real small parameters this branch is real and close to one; the positivity and strict monotonicity of  $\Psi_\varepsilon$  identify it with the desired real root.

Choose a closed polydisc strictly inside the domain of holomorphy, with radii  $R_i$ , on which  $|\mathcal{V}| \leq K$ . Cauchy's coefficient estimate bounds the sum of terms of total degree  $k$  at the stated point by  $K \binom{k+2}{2} \theta^k$ . Summing for  $k > N$ , and using

$$\sum_{k>N} \binom{k+2}{2} \theta^k \leq \frac{(N+3)^2 \theta^{N+1}}{(1-\theta)^3},$$

proves (4.6).

To make the coefficient construction explicit, write  $\mathcal{V} = 1 + \sum_{m \geq 1} P_m$ , where  $P_m$  is homogeneous of degree  $m$ . If  $P_1, \dots, P_{m-1}$  are known, then

$$P_m = -\frac{1}{2}[s^m]\mathcal{F}\left(sr, s\varepsilon, sv; 1 + \sum_{k < m} s^k P_k\right). \quad (4.7)$$

Only finitely many Taylor coefficients enter this extraction. The factor  $1/2$  comes from the same nonzero derivative, so induction proves the recurrence at every degree.  $\square$

## 4.2 The first corrections and their support

The first two homogeneous terms are

$$P_1 = -g_0\varepsilon - \frac{3}{4}v, \quad (4.8)$$

$$P_2 = \left(\frac{3}{2}g_0^2 - \frac{1}{2}g_0 - g_1\right)\varepsilon^2 + \left(\frac{7}{4}g_0 - \frac{3}{8} - g_1 - \frac{1}{2}\gamma_1\right)\varepsilon v + \frac{15}{32}v^2 - \frac{1}{12}rv. \quad (4.9)$$

For example, the endpoint specialization is

$$\frac{t}{r} = 1 - \frac{3}{4}v + \frac{15}{32}v^2 - \frac{1}{12}rv + O((|r| + |v|)^3), \quad \varepsilon = 0. \quad (4.10)$$

The formulas follow by inserting  $V = 1 + P_1 + P_2$  into (4.5) and collecting total degrees. The supplied exact symbolic check verifies that the residual through degree two is identically zero, treating the constants as symbols subject only to  $2g_0 + \gamma = 3/2$ .

The support before evaluation is the ordinary locally finite semigroup  $\mathbb{N}^3$ : terms are  $r^i\varepsilon^jv^k$ . After evaluation,  $v$  is a nontrivial logarithmic/exponential function of  $r$  and  $\varepsilon$ . Thus the theorem gives one convergent analytic chart *over an exact core*; it does not assert that the resulting family has one finite Puiseux or Hahn grid in  $\delta$ , independent of the parameter. That distinction is what permits confluence without introducing a false uniform ramification assertion.

The constants in (4.6) exist by holomorphy; numerical values for a certified polydisc have not been computed here. The separate real inequalities in Section 11 provide finite root brackets without requiring those constants.

## 4.3 Recovery of the two fixed-exponent boundary laws

The same chart also displays the two limits when a sufficiently small nonzero  $\varepsilon$  is held fixed while  $r \downarrow 0$ . For  $\varepsilon > 0$ , one has  $v \rightarrow 0$ , and (4.5) at  $(r, v) = (0, 0)$  reduces to

$$V^{2-\varepsilon}(1 + 2\varepsilon g(\varepsilon)) = 1.$$

For  $\varepsilon < 0$ , one has  $v \rightarrow -\varepsilon$ , and the limiting equation is  $V^2(1 - \varepsilon b(\varepsilon)) = 1$ . Hence, by the branch near one,

$$\mathcal{V}(0, \varepsilon, 0) = [2\varepsilon\Gamma(\varepsilon - 2)]^{-1/(2-\varepsilon)} \quad (\varepsilon > 0), \quad (4.11)$$

$$\mathcal{V}(0, \varepsilon, -\varepsilon) = [-\varepsilon\zeta(1 - \varepsilon)]^{-1/2} \quad (\varepsilon < 0). \quad (4.12)$$

Combining these identities with the core gives the familiar leading forms

$$t \sim \begin{cases} [\delta/(c_\varepsilon\Gamma(\varepsilon - 2))]^{1/(2-\varepsilon)}, & \varepsilon > 0, \\ [2\delta/(c_\varepsilon\zeta(1 - \varepsilon))]^{1/2}, & \varepsilon < 0. \end{cases} \quad (4.13)$$

The negative- $\varepsilon$  side has a finite second moment and a leading square root. It is not an analytic finite-action fold: higher moments and higher boundary derivatives can still be singular.

## 5 The two scales in an arbitrary joint limit

Now let  $\varepsilon_n \rightarrow 0$ , with either sign and without a rate restriction. Write  $\alpha_n = 2 - \varepsilon_n$ ,  $c_n = c_{\varepsilon_n}$ , and define  $A_n, B_n$  by (1.3). All statements below concern sufficiently large  $n$ , so the large root  $B_n > e^2$  is unambiguous. Set

$$v_n = \frac{1}{h_{\varepsilon_n}(B_n)}, \quad d_n = \frac{B_n^{\varepsilon_n}}{h_{\varepsilon_n}(B_n)} = nc_n B_n^{-\alpha_n} = \varepsilon_n + v_n. \quad (5.1)$$

**Lemma 5.1** (Scale separation and retained truncated variance). *The scales satisfy*

$$A_n = n^{1/2+o(1)}, \quad B_n = n^{1/2+o(1)}, \quad v_n, d_n \rightarrow 0, \quad \frac{A_n}{B_n} = d_n^{1/\alpha_n} \rightarrow 0. \quad (5.2)$$

For every fixed  $a > 0$ ,

$$\frac{h_{\varepsilon_n}(aA_n)}{h_{\varepsilon_n}(B_n)} \rightarrow 1, \quad nc_n \sum_{j>aA_n} j^{-\alpha_n} = O_a(A_n) = o(B_n). \quad (5.3)$$

The first assertion is uniform for  $a$  in a compact subset of  $(0, \infty)$ .

*Proof.* The function  $x^2/h_\varepsilon(x)$  is increasing on  $x > e^2$  for small  $|\varepsilon|$ , and tends to infinity, which proves uniqueness and existence of  $B_n$  there. It also shows  $B_n \rightarrow \infty$ . Bound (3.3) then proves  $v_n, d_n \rightarrow 0$ . Moreover

$$h_\varepsilon(x) \leq (\log x) e^{\varepsilon_+ \log x}, \quad \varepsilon_+ = \max(\varepsilon, 0),$$

so  $\log h_{\varepsilon_n}(B_n) = o(\log B_n)$ ; here  $h \rightarrow \infty$ , so no negative lower-order logarithm is needed. Taking logarithms in the defining equation gives  $\log B_n / \log n \rightarrow 1/2$ . The assertion for  $A_n$  is direct, and  $(A_n/B_n)^{\alpha_n} = d_n$  is exact.

For the truncated-variance assertion, put  $B = B_n$ ,  $A = A_n$ ,  $\varepsilon = \varepsilon_n$ ,  $d = d_n$ , and  $\Delta = \log(B/(aA))$ ; eventually  $aA < B$ . If  $\varepsilon \geq 0$ ,

$$\frac{h_\varepsilon(B) - h_\varepsilon(aA)}{h_\varepsilon(B)} = \frac{1}{h_\varepsilon(B)} \int_{aA}^B y^\varepsilon \frac{dy}{y} \leq d\Delta.$$

If  $\varepsilon < 0$ , the same integral is bounded by

$$\frac{(aA)^\varepsilon}{h_\varepsilon(B)} \Delta = a^\varepsilon d^{1+\varepsilon/\alpha_n} \Delta = a^\varepsilon d^{2/\alpha_n} \Delta.$$

Since  $\Delta = -\log a - (\log d)/\alpha_n$ , both bounds tend to zero, uniformly for  $a$  in compact positive intervals. Finally, the integral bound for the decreasing function  $y^{-\alpha_n}$ , together with  $nc_n = A_n^{\alpha_n}$ , bounds the deleted mean by a constant times  $nc_n(aA_n)^{1-\alpha_n} = O_a(A_n)$ .  $\square$

The second-moment sum up to  $B_n$  is of order  $h_{\varepsilon_n}(B_n)$ . Lemma 5.1 says that retaining only actions up to a fixed multiple of  $A_n$ , though  $A_n \ll B_n$ , still retains asymptotically all of this quantity. This fact is stronger than scale separation alone and is the key to the local conditioning theorem.

**Theorem 5.2** (Explicit scale crossover). *Let  $\lambda_n = \varepsilon_n \log n$ . If  $|\lambda_n| \leq K$ , then uniformly for such sequences,*

$$B_n^2 = \frac{c_n}{2} n \log n \mathcal{E}(\lambda_n/2) \left( 1 + O_K \left( \frac{\log \log n}{\log n} \right) \right), \quad (5.4)$$

$$A_n = \sqrt{nc_n} e^{\lambda_n/4} \left( 1 + O_K \left( \frac{1}{\log n} \right) \right). \quad (5.5)$$

In particular, if  $\lambda_n \rightarrow \lambda \in \mathbb{R}$ ,

$$\frac{B_n}{A_n} \sim \sqrt{\log n \frac{1 - e^{-\lambda/2}}{\lambda}}, \quad (5.6)$$

where the fraction at zero equals  $1/2$ . In the outer joint regimes,

$$B_n \sim (nc_n/\varepsilon_n)^{1/(2-\varepsilon_n)} \quad \text{if } \varepsilon_n \log n \rightarrow +\infty, \quad (5.7)$$

$$B_n \sim \sqrt{nc_n/|\varepsilon_n|} \quad \text{if } \varepsilon_n \log n \rightarrow -\infty. \quad (5.8)$$

*Proof.* When  $|\lambda_n| \leq K$ , the defining equation and  $h_\varepsilon(x) = \log x \mathcal{E}(\varepsilon \log x)$  first give  $\log B_n = \frac{1}{2} \log n + O_K(\log \log n)$ . On compact real intervals  $\mathcal{E}$  is positive with bounded logarithmic derivative. Substitution gives (5.4). Expanding  $1/(2 - \varepsilon_n)$  in the exact expression for  $A_n$  gives (5.5). Their quotient gives (5.6).

In the positive outer regime,  $\varepsilon_n \log B_n \rightarrow +\infty$ , by Lemma 5.1. Consequently  $h_{\varepsilon_n}(B_n) \sim B_n^{\varepsilon_n}/\varepsilon_n$ . Raising the resulting relation for  $B_n^{2-\varepsilon_n}$  to the bounded power  $1/(2 - \varepsilon_n)$  proves (5.7). In the negative regime,  $h_{\varepsilon_n}(B_n) \sim 1/|\varepsilon_n|$ , proving (5.8).  $\square$

## 6 A local Gaussian theorem stable under tail thinning

Let  $\varphi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ . In this section the unthinned Poisson means are  $nc_n j^{-1-\alpha_n}$ , so  $\mathbb{E}S_n = n$ .

**Theorem 6.1** (Masked triangular local limit). *Fix  $a > 0$ . For each  $n$ , choose arbitrary numbers  $0 \leq w_{n,j} \leq 1$  satisfying  $w_{n,j} = 1$  for all  $j \leq aA_n$ . Let  $T_n$  be the weighted sum of independent Poisson variables with means  $w_{n,j} nc_n j^{-1-\alpha_n}$ , and let  $\mu_n = \mathbb{E}T_n$ . Then*

$$\sup_{k \in \mathbb{Z}} \left| B_n \Pr(T_n = k) - \varphi \left( \frac{k - \mu_n}{B_n} \right) \right| \rightarrow 0. \quad (6.1)$$

The convergence is uniform over the indicated masks, and  $\mu_n = n + O_a(A_n) = n + o(B_n)$ . In particular,

$$\sup_{k \in \mathbb{Z}} \left| B_n \Pr(S_n = k) - \varphi \left( \frac{k - n}{B_n} \right) \right| \rightarrow 0. \quad (6.2)$$

### 6.1 Convergence on compact Fourier intervals

*Proof of Theorem 6.1, first part.* Suppress the subscript  $n$ , and write  $B = B_n$ ,  $A = A_n$ ,  $\varepsilon = \varepsilon_n$ ,  $\alpha = \alpha_n$ ,  $c = c_n$ ,  $h = h_\varepsilon(B)$ . The centered unthinned logarithmic characteristic function at frequency  $x/B$  is

$$K_n(x) = nc \sum_{j \geq 1} j^{-1-\alpha} \left( e^{ijx/B} - 1 - \frac{ijx}{B} \right). \quad (6.3)$$

On every fixed interval  $|x| \leq T$ , Taylor's formula for  $j \leq B$  gives

$$K_n(x) = -\frac{x^2}{2} \frac{\sum_{j \leq B} j^{1-\alpha}}{h} + O_T(d_n) = -\frac{x^2}{2} + O_T(v_n + d_n). \quad (6.4)$$

Indeed the lower-part cubic error is bounded by

$$C_T \frac{nc}{B^3} \sum_{j \leq B} j^{2-\alpha} \leq C_T nc B^{-\alpha} = C_T d_n.$$

For  $j > B$ , the bound  $|e^{iu} - 1 - iu| \leq 2 + |u|$  gives the same order. The sum-integral estimate (3.4) accounts for  $v_n$ .

The masked centered logarithm has an extra factor  $w_{n,j}$  in the sum. For deleted terms with  $aA < j \leq B$ , the inequality  $|e^{iu} - 1 - iu| \leq u^2/2$  bounds the difference by

$$C_T \frac{\sum_{aA < j \leq B} j^{1-\alpha}}{h}.$$

For  $j > B$ , it is  $O_T(d_n)$ . Lemma 5.1 and (3.4) show that the difference tends to zero, independently of the mask. Thus the centered masked characteristic functions converge to  $e^{-x^2/2}$ , uniformly on compact  $x$ -intervals. The mean estimate is exactly the deleted-mean bound in (5.3).  $\square$

## 6.2 A uniform integrable Fourier majorant

Compact convergence is not enough for a local limit. We next bound the full lattice inversion integral, including frequencies that grow with  $n$ .

**Lemma 6.2** (Two Fourier bounds). *There are positive constants  $C_1, C_2$ , depending at most on  $a$ , such that all sufficiently large  $n$ , and every mask in Theorem 6.1, satisfy*

$$\left| \mathbb{E} e^{ix(T_n - \mu_n)/B_n} \right| \leq e^{-C_1 |x|^{7/4}} \quad (1 \leq |x| \leq \sqrt{B_n}), \quad (6.5)$$

$$\left| \mathbb{E} e^{ix(T_n - \mu_n)/B_n} \right| \leq e^{-C_2 n/B_n} \quad (\sqrt{B_n} \leq |x| \leq \pi B_n). \quad (6.6)$$

*Proof.* The modulus is

$$\exp \left[ -nc \sum_{j \geq 1} w_{n,j} j^{-1-\alpha} (1 - \cos(jx/B)) \right].$$

For the first range put  $y = \min(aA, B/|x|)$ . All integers  $j \leq y$  are retained and satisfy  $|jx/B| \leq 1$ . Since  $1 - \cos u \geq u^2/4$  there,

$$-\log \left| \mathbb{E} e^{ix(T_n - \mu_n)/B} \right| \geq \frac{x^2}{4h} \sum_{j \leq y} j^{1-\alpha} \geq \frac{x^2}{4} \frac{h_\varepsilon(y)}{h}. \quad (6.7)$$

If  $y = aA$ , the ratio tends to one by Lemma 5.1. If  $y = B/|x|$ , write  $L = \log B$ ,  $b = \log |x| \leq L/2$ . For  $\varepsilon \geq 0$ , changing variables in the defining integral shows

$$h_\varepsilon(B/|x|) = e^{-\varepsilon b} \int_b^L e^{\varepsilon u} du \geq \frac{1}{2} e^{-\varepsilon b} h_\varepsilon(B).$$

The last inequality follows because the increasing integrand has at least half its integral on the upper interval of length at least  $L/2$ . For  $\varepsilon < 0$ , the decreasing integrand gives  $\int_0^{L-b} e^{\varepsilon u} du \geq \frac{1}{2} \int_0^L e^{\varepsilon u} du$ . Therefore in either case

$$\frac{h_\varepsilon(B/|x|)}{h_\varepsilon(B)} \geq \frac{1}{2} |x|^{-\varepsilon} \geq \frac{1}{2} |x|^{-1/4}.$$

Together with (6.7) this proves (6.5), with a smaller constant if necessary.

For the second range the retained action  $j = 1$  alone suffices. On  $|u| \leq \pi$ ,  $1 - \cos u \geq 2u^2/\pi^2$ . The exponent is consequently at least  $Cnx^2/B^2$ , which is at least  $Cn/B$  on the stated range. The constants are uniform because  $c_n \rightarrow c_0 > 0$ .  $\square$

*Completion of Theorem 6.1.* Extend the centered scaled characteristic function by zero outside  $[-\pi B_n, \pi B_n]$ . Compact convergence, the integrable majorant in (6.5), and

$$B_n e^{-C_2 n/B_n} \longrightarrow 0$$

from  $B_n = n^{1/2+o(1)}$ , prove convergence in  $L^1(\mathbb{R})$  to  $e^{-x^2/2}$ . Every bound is independent of the mask. Lattice Fourier inversion, after the change of variables  $x = B_n \theta$ , gives

$$B_n \Pr(T_n = k) = \frac{1}{2\pi} \int_{-\pi B_n}^{\pi B_n} e^{-ix(k-\mu_n)/B_n} \mathbb{E} e^{ix(T_n-\mu_n)/B_n} dx.$$

The difference from the Gaussian inversion integral is bounded uniformly in  $k$  by the  $L^1$  difference divided by  $2\pi$ . This proves the claim and completes the proof without an unproved interchange of limits.  $\square$

*Remark 6.3* (Meaning of arbitrary-rate uniformity). The theorem applies to every sequence  $\varepsilon_n \rightarrow 0$ , including sign changes and  $|\varepsilon_n| \log n \rightarrow \infty$ . Equivalently, its errors tend to zero uniformly over any prescribed shrinking neighborhoods  $|\varepsilon| \leq \eta_n \rightarrow 0$ . It does not assert Gaussian convergence uniformly on a fixed exponent neighborhood: a fixed  $\varepsilon > 0$  belongs to a non-Gaussian stable regime.

## 7 Sector asymptotics and sharp action budgets

**Theorem 7.1** (Multiplicative sector equivalent). *For every  $\varepsilon_n \rightarrow 0$ ,*

$$u_n(\varepsilon_n) = \frac{\rho_{\varepsilon_n}^{-n}}{\sqrt{2\pi n B_n}} (1 + o(1)). \quad (7.1)$$

*If  $\varepsilon_n \log n \rightarrow \lambda \in \mathbb{R}$ , then*

$$u_n(\varepsilon_n) \sim \frac{\rho_{\varepsilon_n}^{-n}}{\sqrt{\pi c_0} n^{3/2} \sqrt{\log n} \sqrt{\mathcal{E}(\lambda/2)}}. \quad (7.2)$$

*Proof.* At critical coupling the mean of  $S_n$  is exactly  $n$ . Theorem 6.1 gives  $B_n \Pr(S_n = n) \rightarrow \varphi(0) > 0$ , so its additive local limit is a relative equivalent at that point. Substitute into (2.4), and then use (5.4).  $\square$

**Theorem 7.2** (Cutoff profile and necessary-and-sufficient budget). *Define*

$$H(s) = e^{-1/(2s^2)} \quad (s > 0), \quad H(0) = 0, \quad H(\infty) = 1. \quad (7.3)$$

*For any positive integer sequence  $M_n$ , if  $M_n/A_n \rightarrow s$  in  $[0, \infty]$ , then*

$$R_n(M_n) \longrightarrow H(s). \quad (7.4)$$

*The convergence is uniform when  $M_n/A_n$  ranges over a compact positive interval, with the profile evaluated at the actual ratio. Moreover,*

$$R_n(M_n) \rightarrow 1 \iff M_n/A_n \rightarrow \infty. \quad (7.5)$$

*For fixed allowed loss  $\theta \in (0, 1)$ , let  $M_{\min}(n, \theta) = \min\{M : R_n(M) \geq 1 - \theta\}$ . Then*

$$\frac{M_{\min}(n, \theta)}{A_n} \longrightarrow \frac{1}{\sqrt{2 \log(1/(1 - \theta))}}. \quad (7.6)$$

*Proof.* First suppose  $M_n/A_n \rightarrow s \in (0, \infty)$ . An integral comparison, uniform for  $\alpha_n$  near two, gives

$$\Lambda_{n,M_n} = nc_n \sum_{j>M_n} j^{-1-\alpha_n} = \frac{nc_n}{\alpha_n M_n^{\alpha_n}} (1 + O(M_n^{-1})) \rightarrow \frac{1}{2s^2}. \quad (7.7)$$

The cutoff mask retains every action up to  $aA_n$ , for any fixed  $0 < a < s$  and all sufficiently large  $n$ . Its mean differs from  $n$  by  $O(A_n) = o(B_n)$ . The masked local limit thus gives  $B_n \Pr(S_{n,M_n} = n) \rightarrow \varphi(0)$ , as for the full sum. The exact ratio (2.5) proves (7.4).

Uniformity on compact positive intervals follows either from the uniform mask bounds or from the usual monotonicity argument: choose a finite mesh, apply pointwise convergence on that mesh, and use uniform continuity of the continuous increasing function  $H$  between adjacent mesh points.

If  $M_n/A_n \rightarrow 0$ , compare  $M_n$  to  $\lfloor sA_n \rfloor$  for every fixed  $s > 0$ , and then let  $s \downarrow 0$ . If  $M_n/A_n \rightarrow \infty$ , compare in the opposite direction and let  $s \rightarrow \infty$ . This proves the endpoint statements. Conversely, if the ratio of cutoffs does not tend to infinity, some subsequence is bounded above by a fixed  $C$ . On that subsequence  $R_n(M_n) \leq R_n(\lceil CA_n \rceil) \rightarrow H(C) < 1$ , proving necessity in (7.5).

Finally, the strictly increasing continuous profile crosses  $1 - \theta$  only at the value displayed in (7.6). Compare cutoffs just below and just above that value times  $A_n$ , and use monotonicity and  $A_n \rightarrow \infty$  to remove integer rounding.  $\square$

**Corollary 7.3** (The explicit exponent/order window). *If  $\varepsilon_n \log n \rightarrow \lambda \in \mathbb{R}$  and  $M_n/\sqrt{nc_0} \rightarrow m > 0$ , then*

$$R_n(M_n) \rightarrow \exp\left(-\frac{e^{\lambda/2}}{2m^2}\right). \quad (7.8)$$

*Thus a budget appropriate to the endpoint  $\lambda = 0$  does not retain the same fraction throughout the crossover window.*

*Proof.* Equation (5.5) gives  $M_n/A_n \rightarrow me^{-\lambda/4}$ . Apply Theorem 7.2.  $\square$

These are statements about a *prefix cutoff*. No optimality among all possible non-prefix subsets of actions is claimed. Likewise, (7.6) is proved for a fixed loss tolerance, not for a tolerance  $\theta_n$  tending to zero. The latter requires quantitative errors.

## 8 Conditional Poisson limits for the entire extreme-action process

Define the random point measure

$$\Pi_n = \sum_{j \geq 1} Z_{n,j} \delta_{j/A_n} \quad \text{on } (0, \infty). \quad (8.1)$$

It is locally finite. A Poisson point process with intensity  $y^{-3} dy$  is characterized by the Laplace functional

$$\exp\left[-\int_0^\infty (1 - e^{-f(y)}) y^{-3} dy\right], \quad f \geq 0, \quad f \in C_c((0, \infty)). \quad (8.2)$$

The infinite intensity near zero causes no problem for compactly supported test functions in  $(0, \infty)$ .

**Theorem 8.1** (Local factorization under exact conditioning). *Let  $\varepsilon_n \rightarrow 0$ , and let integers  $k_n$  satisfy  $(k_n - n)/B_n \rightarrow x \in \mathbb{R}$ . For every nonnegative  $f \in C_c((0, \infty))$ ,*

$$\begin{aligned} B_n \mathbb{E} \left[ e^{-\Pi_n(f)} \mathbf{1}_{\{S_n = k_n\}} \right] \\ \longrightarrow \varphi(x) \exp \left[ - \int_0^\infty (1 - e^{-f(y)}) y^{-3} dy \right]. \end{aligned} \quad (8.3)$$

*Consequently, under  $S_n = k_n$ , the point process  $\Pi_n$  converges to that Poisson point process. The limit is independent of the limiting central location  $x$ . Unconditionally,  $((S_n - n)/B_n, \Pi_n)$  converges jointly to an independent standard normal variable and the same Poisson process.*

*Proof.* Choose  $a > 0$  below the support of  $f$ , and set  $w_{n,j} = e^{-f(j/A_n)}$ . Poisson exponential tilting, first for finitely many nontrivial weights and then directly for the present compact support, gives the exact identity

$$\mathbb{E}[e^{-\Pi_n(f)} \mathbf{1}_{\{S_n = k\}}] = e^{-L_n(f)} \Pr(T_n = k), \quad L_n(f) = \sum_j n c_n j^{-1-\alpha_n} (1 - w_{n,j}), \quad (8.4)$$

where  $T_n$  uses the masked means from Theorem 6.1. The identity follows already for one Poisson variable from multiplying its mass at  $m$  by  $w^m$ ; independence gives the product.

Since  $n c_n A_n^{-\alpha_n} = 1$ , a Riemann sum on the compact support of  $f$ , bounded away from zero, yields

$$L_n(f) = \frac{1}{A_n} \sum_j (j/A_n)^{-1-\alpha_n} (1 - e^{-f(j/A_n)}) \longrightarrow \int_0^\infty (1 - e^{-f(y)}) y^{-3} dy. \quad (8.5)$$

The masked mean is  $n + O(A_n) = n + o(B_n)$ . Its local limit therefore gives  $B_n \Pr(T_n = k_n) \rightarrow \varphi(x)$ . Combine this with (8.4) to obtain (8.3). Dividing by  $B_n \Pr(S_n = k_n) \rightarrow \varphi(x) > 0$  proves convergence of the conditional Laplace functionals, which characterizes the point-process limit.

For the unconditional joint assertion, the same proof is uniform for  $(k - n)/B_n$  in a fixed compact interval. Sum (8.4) against a bounded continuous function of  $(k - n)/B_n$ ; the local limit gives a Gaussian Riemann integral on that interval. The unmasked central limit, already implied by compact characteristic-function convergence, controls the complement. This proves factorization for products of a central test function and a point-process Laplace functional, and hence the stated joint convergence.  $\square$

The local assertion is stronger than an unconditional independence statement: the event  $S_n = k_n$  has probability of order  $1/B_n$ . An unconditional weak limit alone would not justify dividing by that vanishing probability. The masked local limit is precisely what makes that division legitimate.

**Corollary 8.2** (Ordered extreme actions). *Let  $J_{n,(1)} \geq J_{n,(2)} \geq \dots$  be the action sizes in the Poisson multiset, including multiplicities and padded by zeros if necessary. Under  $S_n = n$ , for each fixed  $r \geq 1$  and  $s > 0$ ,*

$$\Pr \left( J_{n,(r)}/A_n \leq s \mid S_n = n \right) \longrightarrow e^{-\mu(s)} \sum_{\ell=0}^{r-1} \frac{\mu(s)^\ell}{\ell!}, \quad \mu(s) = \frac{1}{2s^2}. \quad (8.6)$$

*Proof.* On any bounded interval bounded away from zero, Theorem 8.1 gives the corresponding Poisson counting limit; boundary points have zero limiting intensity. To pass from  $(s, R]$  to  $(s, \infty)$ , observe from Theorem 7.2 that the conditional probability of any action exceeding  $RA_n$  tends to  $1 - H(R)$ , which vanishes as  $R \rightarrow \infty$ . The limiting number above  $s$  is Poisson of mean  $\int_s^\infty y^{-3} dy = 1/(2s^2)$ . The  $r$ -th largest is at most  $s$  precisely when this count is at most  $r - 1$ .  $\square$

## 9 A moving coupling window and finite-prefix robustness

**Theorem 9.1** (A bounded critical coupling window). *Let  $\varepsilon_n \rightarrow 0$ , and let  $\omega_n$  range over a fixed compact subset of  $\mathbb{R}$ . Define*

$$\beta_n = 1 + \omega_n \frac{B_n}{n}. \quad (9.1)$$

*Then  $\beta_n > 0$  for all sufficiently large  $n$ , and uniformly over this window,*

$$u_n(\varepsilon_n, \beta_n) = \frac{e^{n\beta_n F_{\varepsilon_n}(1)}}{\sqrt{2\pi} n B_n} e^{-\omega_n^2/2} (1 + o(1)). \quad (9.2)$$

*For  $M_n/A_n \rightarrow s > 0$ , the cutoff ratio remains  $H(s)$ . Conditioned on the total weighted action being  $n$ , the point-process limit remains the Poisson process with intensity  $y^{-3} dy$ .*

*Proof.* The factor  $\beta_n \rightarrow 1$  multiplies every Poisson mean. The centered characteristic exponent and each Fourier lower bound in the proof of Theorem 6.1 are multiplied by this factor. All compact limits and integrable bounds therefore remain valid uniformly in  $\omega_n$ . The unmasked mean is  $n\beta_n = n + \omega_n B_n$ . At the target lattice point  $n$ , the centered coordinate is exactly  $-\omega_n$ , giving the local factor  $\varphi(\omega_n)$ , bounded away from zero on the compact window. Equation (2.4) proves (9.2).

Deletion above a fixed multiple of  $A_n$  still shifts the mean by only  $O(A_n) = o(B_n)$ . Thus numerator and denominator in the cutoff ratio have the same Gaussian factor. The deleted intensity has the same limit because  $\beta_n \rightarrow 1$ . Similarly, the tilted Riemann sum and the ratio of local masses in (8.4) are unchanged in the limit, proving the point-process assertion.  $\square$

For  $\beta_n \neq 1$ , the carrier  $e^{n\beta_n F_{\varepsilon_n}(1)}$  in (9.2) is an exact factor from coefficient extraction. It is not being identified with the reciprocal radius of convergence raised to the  $n$ -th power. In particular, on the supercritical side the dominant singularity can move into the analytic interior. The bounded-window proof does not require a separate assertion about that singularity.

**Corollary 9.2** (Robustness under a bounded finite prefix). *The probabilistic results of Sections 6–9 remain valid for weights  $a_{n,j} \geq 0$  satisfying, for some fixed  $J$ ,*

$$a_{n,j} = c_n j^{-3+\varepsilon_n} \quad (j > J), \quad \sum_{j \geq 1} j a_{n,j} = 1, \quad (9.3)$$

*where  $\varepsilon_n \rightarrow 0$ ,  $c_n \rightarrow c_* > 0$ , the finitely many prefix weights are bounded, and  $a_{n,1} \geq a_* > 0$ . Define  $A_n, B_n$  using this  $c_n$ , and set  $\rho_n = \exp(-\sum_j a_{n,j})$ . In the explicit bounded-window formulas replace  $c_0$  by  $c_*$ .*

*Proof.* The scale lemmas use only  $c_n \rightarrow c_* > 0$ . After exact centering, altering a fixed bounded prefix changes the compact characteristic exponent by  $O(n/B_n^2) = O(1/h_{\varepsilon_n}(B_n)) = o(1)$ . The tail integrals, mean deletion bounds, and Riemann sums are unchanged. In the near-frequency bound, omitting finitely many terms changes the diverging second-moment sum by  $O(1)$ , which is negligible even at the smallest truncation used there. The far bound uses  $a_{n,1} \geq a_* > 0$  instead of  $c_n$ . The Lagrange and Poisson identities hold for arbitrary nonnegative weights, completing the proof.  $\square$

This corollary concerns the asymptotic coefficient and point-process results. The explicit chart coefficients (4.8)–(4.9) belong to the normalized polylogarithm model; finite-prefix changes generally alter them.

## 10 Inverse crossover profiles and an exact core duality

**Theorem 10.1** (Real inverse crossover). *Let  $\delta \downarrow 0$ ,  $\varepsilon \rightarrow 0$ , and let  $t > 0$  solve  $\Psi_\varepsilon(t) = \delta$ . Write  $L = \log(1/\delta)$ . Uniformly when  $|\varepsilon L| \leq K$ ,*

$$t = 2\sqrt{\frac{\delta}{c_\varepsilon L \mathcal{E}(\varepsilon L/2)}} \left(1 + O_K\left(\frac{\log L}{L}\right)\right). \quad (10.1)$$

In the outer joint regimes,

$$t \sim (2\varepsilon\delta/c_\varepsilon)^{1/(2-\varepsilon)} \quad \text{if } \varepsilon L \rightarrow +\infty, \quad (10.2)$$

$$t \sim \sqrt{2|\varepsilon|\delta/c_\varepsilon} \quad \text{if } \varepsilon L \rightarrow -\infty. \quad (10.3)$$

At  $\varepsilon = 0$ , the leading form is  $t \sim 2\sqrt{\delta/(c_0 \log(1/\delta))}$ . In all these limits,  $U_{\varepsilon,c} - U_\varepsilon(\rho_\varepsilon e^{-\delta}) \sim t$ .

*Proof.* The chart gives  $t/r \rightarrow 1$  in every joint limit. Put  $\ell = \log(1/r)$ . Taking logarithms of the core equation shows  $\ell \sim L/2$ , by the same estimates used for  $\log B_n$ . If  $|\varepsilon L| \leq K$ , refine this to  $\ell = L/2 + O_K(\log L)$ . Hence

$$h_\varepsilon(1/r) = \frac{L}{2} \mathcal{E}(\varepsilon L/2) \left(1 + O_K\left(\frac{\log L}{L}\right)\right).$$

The core equation gives the displayed formula for  $r$ ; the chart error is  $O(|\varepsilon| + v + r) = O_K(1/L)$ , which is smaller than the retained error.

When  $\varepsilon L \rightarrow +\infty$ , one has  $h_\varepsilon(1/r) \sim r^{-\varepsilon}/\varepsilon$ ; on the negative side it is  $\sim 1/|\varepsilon|$ . Solve the core equation in each case and use  $t/r \rightarrow 1$ . Finally,  $\delta/t = \Psi_\varepsilon(t)/t \rightarrow 0$  uniformly in this joint limit: (3.5) and  $h_\varepsilon(1/t) \leq \log(1/t)t^{-\varepsilon+}$  make this immediate for  $|\varepsilon| < 1/4$ . Apply (2.9).  $\square$

**Proposition 10.2** (Exact core duality and boundary/sector matching). *For all sufficiently large  $n$ , the two core normalizations satisfy the exact identity*

$$r_{\varepsilon_n} \left(\frac{1}{2n}\right) = \frac{1}{B_n}. \quad (10.4)$$

Moreover,

$$\frac{n\rho_{\varepsilon_n}^n u_n(\varepsilon_n)}{U_{\varepsilon_n,c} - U_{\varepsilon_n}(\rho_{\varepsilon_n} e^{-1/(2n)})} \rightarrow \frac{1}{\sqrt{2\pi}}. \quad (10.5)$$

*Proof.* Substituting  $r = 1/B_n$  into the core equation gives

$$\frac{c_n}{2B_n^2} h_{\varepsilon_n}(B_n) = \frac{1}{2n},$$

so uniqueness proves (10.4). The convergent chart gives  $t(1/(2n)) \sim 1/B_n$ . Also  $B_n/n \rightarrow 0$ , so the term  $\delta = 1/(2n)$  is negligible in  $t - \delta$ . The numerator in (10.5) equals  $\Pr(S_n = n) \sim 1/(\sqrt{2\pi}B_n)$ , which proves the result.  $\square$

The exact identity (10.4) explains why the same crossover factor appears in local inversion and large-sector coefficients. Equation (10.5) is a leading-order bridge, not an all-orders transfer theorem. Constructing a uniform correction calculus on the Fourier side is a distinct next step.

## 11 Finite real bounds for inversion

The convergent chart has an existence-level geometric remainder. The following finite inequalities offer an independent route to real enclosures. They also avoid evaluating nearly cancelling gamma and zeta poles.

**Proposition 11.1** (Finite inversion inequalities). *For real  $|\varepsilon| < 1/4$ ,  $\alpha = 2 - \varepsilon$ , integer  $J \geq 1$ , and  $t > 0$ , set*

$$H_J = \sum_{j=1}^J j^{1-\alpha}, \quad K_J = \sum_{j=1}^J j^{2-\alpha}. \quad (11.1)$$

Then

$$\begin{aligned} L_J(t) &:= c_\varepsilon \left( \frac{t^2 H_J}{2} - \frac{t^3 K_J}{6} \right) \leq \Psi_\varepsilon(t) \\ &\leq c_\varepsilon \left( \frac{t^2 H_J}{2} + \frac{t J^{1-\alpha}}{\alpha - 1} \right) =: U_J(t), \end{aligned} \quad (11.2)$$

$$c_\varepsilon \left( t H_J - \frac{t^2 K_J}{2} \right) \leq \Psi'_\varepsilon(t) \leq c_\varepsilon \left( t H_J + \frac{J^{1-\alpha}}{\alpha - 1} \right). \quad (11.3)$$

If  $0 < t_- < t_+$ , and finite evaluations establish

$$U_{J-}(t_-) \leq \delta \leq L_{J+}(t_+), \quad (11.4)$$

then the root of  $\Psi_\varepsilon(t) = \delta$  lies in  $[t_-, t_+]$ .

*Proof.* For all  $x \geq 0$ , Taylor's integral remainder gives

$$\frac{x^2}{2} - \frac{x^3}{6} \leq e^{-x} - 1 + x \leq \frac{x^2}{2}, \quad 0 \leq e^{-x} - 1 + x \leq x,$$

and  $x - x^2/2 \leq 1 - e^{-x} \leq x$ , together with  $1 - e^{-x} \leq 1$ . Apply the quadratic or cubic bounds to  $j \leq J$ , keep the omitted part nonnegative for the lower bounds, and use

$$\sum_{j>J} j^{-\alpha} \leq \int_J^\infty y^{-\alpha} dy = \frac{J^{1-\alpha}}{\alpha - 1}$$

for the upper tails. This proves both inequalities. Strict monotonicity of  $\Psi_\varepsilon$  proves the bracket assertion.  $\square$

If  $t_0 \in [t_-, t_+]$  and a positive lower derivative bound  $m$  is obtained at  $t_-$  from (11.3), then convexity makes  $m$  a lower derivative bound on the whole interval. Therefore

$$|t - t_0| \leq \frac{\max\{|L_J(t_0) - \delta|, |U_J(t_0) - \delta|\}}{m}. \quad (11.5)$$

Choosing  $J$  of order  $1/t$  is natural near the boundary; larger or adaptively chosen finite sums can sharpen the bracket. These statements are rigorous inequalities between real numbers. To turn them into numerical certificates, the finite sums, powers, and  $c_\varepsilon$  must themselves be outward-rounded. The supplied numerical diagnostics do not do that, and are not labeled interval certificates.

## 12 Reproducible checks and numerical diagnostics

The package contains `code/verify.py`, exact-check reports, CSV data, and the generated tables below. The analytic proofs do not depend on the numerical observations. The checks serve different purposes and are reported separately rather than combined into a misleading “verified theorem” label.

### 12.1 Exact finite algebra

The program uses rational arithmetic to compare direct formal fixed-point iteration with the Lagrange formula for

$$U(q) = \sum_{1 \leq j \leq M} \frac{q^j e^{jU(q)}}{j^3}.$$

This rational-amplitude test model deliberately omits the critical factor  $c_0 = 1/\zeta(2)$ . It checks the general algebraic identity, not a claim that the critically normalized coefficients are rational. Degrees 1 through 12, five cutoffs, positivity, and equality of coefficients below the cutoff are checked exactly. A symbolic residual check independently verifies (4.8)–(4.9). The combined report records **122 passing exact assertions**. These are finite computer-algebra checks, not a proof-assistant formalization of the infinite statements.

### 12.2 Positive coefficient recurrences

For Poisson means  $\nu_j$ , define  $p_k = \Pr(S = k)$ . Differentiation of the probability generating function gives the positive recurrence

$$p_0 = e^{-\sum_j \nu_j}, \quad kp_k = \sum_{j=1}^k j\nu_j p_{k-j}. \quad (12.1)$$

The total intensity includes the full infinite tail even though only  $j \leq k$  enters the recurrence for  $p_k$ . For a cutoff the intensity and recurrence are both truncated consistently. The program evaluates this recurrence through  $n = 8192$ , and forms the ratio using the exact expression (2.5), with high-precision evaluations of the tail intensity.

| $n$  | $\lambda$ | $A_n/B_n$ | $\sqrt{2\pi}B_n \Pr(S_n = n)$ | $R_n(A_n)$ | $R_n(1.5A_n)$ |
|------|-----------|-----------|-------------------------------|------------|---------------|
| 512  | -1        | 0.49813   | 0.83518                       | 0.63957    | 0.87800       |
| 512  | 0         | 0.53482   | 0.81565                       | 0.61883    | 0.85704       |
| 512  | 1         | 0.57793   | 0.76690                       | 0.58099    | 0.83451       |
| 2048 | -1        | 0.44669   | 0.86360                       | 0.65661    | 0.86320       |
| 2048 | 0         | 0.48270   | 0.84967                       | 0.63216    | 0.85032       |
| 2048 | 1         | 0.52348   | 0.81450                       | 0.59662    | 0.83748       |
| 8192 | -1        | 0.40890   | 0.88359                       | 0.64122    | 0.85878       |
| 8192 | 0         | 0.44420   | 0.87327                       | 0.62845    | 0.84946       |
| 8192 | 1         | 0.48330   | 0.84665                       | 0.60452    | 0.83788       |

Table 1: Coefficient diagnostics for  $\varepsilon = \lambda/\log n$ . Cutoff arguments are rounded down to integers. The limiting values in the last three columns are 1,  $e^{-1/2} \approx 0.60653$ , and  $e^{-2/9} \approx 0.80074$ , respectively.

The logarithmic convergence is slow, and the cutoff rounding is visible; the table is not evidence for a monotone error law. Six separate comparisons at  $n = 128$ , using an independent arbitrary-precision implementation of (12.1), gave maximum relative disagreement  $7.13 \times 10^{-18}$  or less. That tests the finite numerical calculation, not its distance from the asymptotic limit.

The large-order recurrence uses NumPy’s extended-range `longdouble` when available. The program detects platforms with insufficient exponent range and raises an error instead of silently underflowing. The exact checks and the arbitrary-precision inverse checks do not require that extended floating-point format. None of these calculations is outward-rounded.

### 12.3 Direct checks of the convergent chart

Choose  $r = e^{-L_r}$ ,  $\varepsilon = \tau/L_r$ , and set  $\delta$  by the exact core. The program solves the original polylogarithm equation at 160 decimal digits and compares  $t/r$  with  $1 + P_1$  and  $1 + P_1 + P_2$ . All nine root residuals, measured as  $|\Psi_\varepsilon(t)/\delta - 1|$ , are below  $10^{-55}$ .

| $\log(1/r)$ | $\varepsilon \log(1/r)$ | $t/r$       | degree-1 error         | degree-2 error         |
|-------------|-------------------------|-------------|------------------------|------------------------|
| 20          | -2                      | 0.962316330 | $2.916 \times 10^{-3}$ | $2.837 \times 10^{-4}$ |
| 20          | 0                       | 0.963656348 | $1.156 \times 10^{-3}$ | $1.553 \times 10^{-5}$ |
| 20          | 2                       | 0.933247764 | $8.874 \times 10^{-3}$ | $2.230 \times 10^{-4}$ |
| 40          | -2                      | 0.980462618 | $7.624 \times 10^{-4}$ | $3.750 \times 10^{-5}$ |
| 40          | 0                       | 0.981541081 | $2.911 \times 10^{-4}$ | $1.888 \times 10^{-6}$ |
| 40          | 2                       | 0.968813278 | $2.248 \times 10^{-3}$ | $2.660 \times 10^{-5}$ |
| 80          | -2                      | 0.990045254 | $1.952 \times 10^{-4}$ | $4.824 \times 10^{-6}$ |
| 80          | 0                       | 0.990698010 | $7.301 \times 10^{-5}$ | $2.325 \times 10^{-7}$ |
| 80          | 2                       | 0.984965161 | $5.653 \times 10^{-4}$ | $3.246 \times 10^{-6}$ |

Table 2: Direct inversion diagnostics. The error columns are absolute errors in the ratio  $t/r$ , not in  $t$ . The two approximations are the Taylor polynomials by total degree from (4.8)–(4.9).

The observed improvement is consistent with the convergent chart but does not supply a numerical value for the Cauchy constants in (4.6). The reports distinguish a small evaluated residual from an interval enclosure of the true root.

### 12.4 Rebuilding and rerunning

The source is self-contained apart from standard TeX packages and the two included table fragments. From the package directory, run

```
python code/verify.py --part all
latexmk -pdf -interaction=nonstopmode Confluent_Critical_Transseries.tex
```

The options `-part exact`, `-part coefficients`, and `-part inversion` run the groups independently. Recorded package versions, build instructions, and validation status are included alongside the article. Python assertions should not be disabled with optimization flags when running the check program.

## 13 Further research questions and concrete targets

**1. Uniform higher-order sector corrections.** The local chart is convergent to all orders, whereas the triangular sector law is leading-order. Expand the centered characteristic exponent beyond (6.4) in a set of parameters that remains ordered when  $\varepsilon \log n$  crosses zero. The terms generated by  $x^{2-\varepsilon}$ ,  $x^2$ , and their coalescing coefficients should be retained in cancellation-free blocks. A useful target is a relative expansion uniform for  $|\varepsilon \log n| \leq K$ , with an explicit first correction and a remainder small enough to distinguish it from the leading Gaussian term.

**2. Quantitative cutoff errors and shrinking tolerances.** Make the compact characteristic-function error and the two tail integrals explicit in  $v_n, d_n$ , and in  $1 - h_{\varepsilon_n}(sA_n)/h_{\varepsilon_n}(B_n)$ . This would give a computable error for  $R_n(M)$  on compact positive cutoff intervals. Then study  $\theta_n \downarrow 0$  in (7.6). The fixed-tolerance quantile argument does not itself justify a vanishing-tolerance budget. An eventual interval implementation should certify a finite coefficient ratio, not merely print its limiting profile.

**3. The largest coupling window with untilted extremes.** In Theorem 9.1, a bounded central displacement has no effect on the limiting extreme process. If the target displacement is  $x_n B_n$ , a heuristic Gaussian density ratio for adding an action of size  $A_n y$  contains  $\exp(x_n A_n y / B_n)$ . This suggests that a new tilt can emerge when  $|x_n| A_n / B_n$  is order one. Determine the actual validity range of this heuristic, the necessary moderate-deviation estimates, and the resulting conditional point process. The heuristic is not a theorem of this article.

**4. General triangular slowly varying tails.** Replace the exact tail by  $c_n j^{-3+\varepsilon_n} \ell_n(j)$ . Identify verifiable uniform conditions under which an implicit truncated-second-moment scale and a tail-quantile extreme scale still satisfy the retained-variance property (5.3). Mere slow variation separately for each  $n$  is unlikely to control the joint limit. The masked local-limit proof suggests that this retained-variance property, an aperiodicity condition, and uniform tail bounds are the correct inputs for a more general theorem.

**5. Joint finite-fold drift.** Truncating the kernel restores an analytic finite-action fold. Determine the critical-point and critical-value drift uniformly in  $(\varepsilon, M, n)$ , particularly for  $M \asymp A_n$ . Match the resulting saddle formula to the cutoff factor  $H(M/A_n)$  with an error that remains small after multiplication of logarithmic errors by  $n$ . This would extend the repository's fixed-exponent finite-fold analysis into the confluent window.

**6. Complex continuation of the core.** The analytic correction  $\mathcal{V}$  is holomorphic in independent local variables, but the parameterized core  $r_\varepsilon(\delta)$  has its own complex branch geometry. Construct explicit sectorial domains and track their critical values uniformly as  $\varepsilon \rightarrow 0$ . Only after doing so is it reasonable to ask for uniform complex asymptotics, Stokes transitions, or resurgent properties. A convergent correction chart alone does not establish any of those global assertions.

**7. Nonlinear slopes.** For feedback of the form  $U = \sum a_j q^j \exp((j + r_j)U)$ , the substitution  $z = qe^U$  no longer reduces the problem to a scalar polylogarithm. Start with  $r_j = \gamma j^\theta$ ,  $0 < \theta < 1$ , and determine whether it changes the extreme scale, the Gaussian normalization, or only the critical centering. A successful result must replace the exact coefficient/Poisson identity, not use it after its hypotheses have been lost.

**8. Signed weights and competing singularities.** Positivity underlies cutoff monotonicity, the conditioning interpretation, and several estimates. For signed or complex amplitudes, seek explicit analytic hypotheses that permit a multi-saddle replacement. A particularly concrete target is a finite signed prefix with a positive tail, where the local normal form still exists but coefficient cancellation can invalidate probability arguments.

**9. Growing prefixes and the edge of robustness.** Corollary 9.2 keeps the altered prefix length fixed. Permit a prefix length  $J_n \rightarrow \infty$ , and characterize the allowable changes by their centered second-moment contribution and their action-scale mass. Perturbations at  $j \asymp A_n$

can change the limiting Poisson intensity while leaving the central Gaussian limit unchanged; perturbations spread through the variance-carrying range can change both. Prove a classification that separates these effects quantitatively.

**10. The lower endpoint  $\alpha \downarrow 1$ .** The other half of the repository’s endpoint question remains distinct. There the first moment, not just the second, approaches divergence; the present normalization  $c_\varepsilon = 1/\zeta(\alpha)$  tends to zero and the centering mechanism changes. Derive an appropriate compensated core and determine whether conditioning couples the largest actions to the total at leading order. None of the upper-endpoint Gaussian estimates should be transported there without a new proof.

**11. Uniform matching to the fixed stable conditioned process.** For fixed  $\alpha < 2$ , the natural fluctuation and largest-action scales are comparable, while they separate in the joint limit proved here. Construct an explicit normalization of the fixed stable bridge and show how its conditioned jump process tends to the independent Poisson extreme process as  $\alpha \uparrow 2$ . Then compare that limit with the present arbitrary-rate theorem. This would explain the transition at the level of entire conditioned processes, not only coefficient ratios.

**12. Proof-producing computation and formalization.** A practical first milestone is to formalize the finite identities (2.3), (2.5), and (11.2), while keeping their different algebraic and analytic hypotheses explicit. Next build an interval evaluator for the real core and finite root brackets. The harder Lean interfaces concern the masked Fourier majorant, countable Poisson sums, and local conditioning. An implementation should expose separately: an exact coefficient identity, a certified finite numerical bound, and an asymptotic theorem with its hypotheses.

## 14 Conclusion

The upper endpoint is not obtained by inserting  $\alpha = 2$  into a fractional-power formula. The divergent gamma and zeta terms must first be combined. Their exact combination is the elementary function  $h_\varepsilon$ , which produces both a convergent inverse chart and the correct triangular fluctuation scale.

For every rate of approach to exponent two, the extreme scale becomes small relative to the fluctuation scale, yet a fixed multiple of the extreme scale retains asymptotically all of the relevant truncated variance. This is why exact conditioning preserves a Poisson extreme process and why the action budget has a sharp explicit profile. The same normalization enters the local inverse through the exact identity  $r_\varepsilon(1/(2n)) = 1/B_n$ .

These results resolve a specified upper-endpoint model problem left open in the inspected research sequence. They supply new material for that sequence without treating the fixed-endpoint classical mechanism as a new discovery, without promoting numerical agreement to proof, and without conflating a convergent local chart with global transseries realization or resurgence.

## A Source audit and dependency boundaries

### A.1 Repository and library provenance

The repository snapshot used for the final open-question crosswalk is

0c973d8f5e7ef4550f4df1bf486d890037fd9f14.

The inspected project group is

`Analysis/Transseries/docs/series-and-transseries/.`

The project README, group index, and the critical-Hahn article’s research questions and dependency discussion were read. The exact upper-endpoint question was checked in the pinned article at source lines 1467–1472. The full canonical transseries volume and every unmerged arrival were not audited. Accordingly, this article makes a precise continuation claim for the identified question, not an assertion that all possible repository overlap has been excluded.

The user’s Library supplied the scope statements of `article(20260929-201315).tex`, titled *The Logarithmic Critical Endpoint*, and `Marginal_Critical_Transseries.tex`. These documents identify the fixed endpoint and the still-unaddressed uniform crossover. They are unpublished research drafts, not independent validation of the present claims. No repository files were modified, and no Lean build of this article’s theorems was performed.

## A.2 Proof dependencies

The analytic branch of the argument is

polylogarithm expansion  $\implies$  pole cancellation  $\implies$  analytic implicit chart.

The coefficient branch is

Lagrange identity  $\implies$  Poisson representation  $\implies$  masked local limit.

The scale estimates needed by the second branch follow directly from the integral defining  $h_\varepsilon$ , not from an assumed stable-limit theorem. Poisson tilting and tail Riemann sums then give the conditional process; positivity and monotonicity give the sharp budget and its necessity. The exact core duality connects the two branches after their independent construction. The finite root bounds use only scalar exponential inequalities and integral comparisons. All computational checks are downstream of these proofs.

## B A notation and regime ledger

| Symbol                                 | Meaning                                                                                    |
|----------------------------------------|--------------------------------------------------------------------------------------------|
| $\varepsilon, \alpha$                  | Moving parameter and exponent, related by $\alpha = 2 - \varepsilon$ .                     |
| $c_\varepsilon, F_\varepsilon$         | Critical normalization $1/\zeta(2 - \varepsilon)$ and normalized polylogarithm kernel.     |
| $\rho_\varepsilon, U_{\varepsilon, c}$ | Critical radius $e^{-F_\varepsilon(1)}$ and boundary value $F_\varepsilon(1)$ .            |
| $\delta, t$                            | Boundary coordinates $q = \rho_\varepsilon e^{-\delta}$ , $z = e^{-t}$ .                   |
| $h_\varepsilon, \mathcal{E}$           | Exact crossover integral and entire divided exponential.                                   |
| $r, v, \mathcal{V}$                    | Exact inverse core, its inverse-logarithmic parameter, and convergent analytic correction. |
| $A_n$                                  | Extreme-action scale $(nc_n)^{1/\alpha_n}$ .                                               |
| $B_n$                                  | Large root of $B_n^2 = nc_n h_{\varepsilon_n}(B_n)$ ; central fluctuation scale.           |
| $v_n, d_n$                             | Small quantities $h_{\varepsilon_n}(B_n)^{-1}$ and $nc_n B_n^{-\alpha_n}$ .                |
| $S_n, T_n$                             | Full and thinned compound Poisson weighted sums.                                           |
| $R_n(M), H(s)$                         | Exact retained coefficient fraction and limiting profile $e^{-1/(2s^2)}$ .                 |
| $\Pi_n$                                | Point process of action sizes divided by $A_n$ , with multiplicities.                      |
| $\lambda_n$                            | Coefficient crossover parameter $\varepsilon_n \log n$ .                                   |
| $\omega_n$                             | Bounded coupling displacement in $\beta_n = 1 + \omega_n B_n/n$ .                          |

| Joint regime                                            | Fluctuation normalization                          | Extreme normalization                  |
|---------------------------------------------------------|----------------------------------------------------|----------------------------------------|
| $\varepsilon \log n \rightarrow \lambda \in \mathbb{R}$ | $\sqrt{(c_0/2)n \log n \mathcal{E}(\lambda/2)}$    | $\sqrt{nc_0} e^{\lambda/4}$            |
| $\varepsilon \log n \rightarrow +\infty$                | $(nc_\varepsilon/\varepsilon)^{1/(2-\varepsilon)}$ | $(nc_\varepsilon)^{1/(2-\varepsilon)}$ |
| $\varepsilon \log n \rightarrow -\infty$                | $\sqrt{nc_\varepsilon/ \varepsilon }$              | $(nc_\varepsilon)^{1/(2-\varepsilon)}$ |

In every row  $\varepsilon \rightarrow 0$  is part of the hypothesis. The entries for  $B_n$  are asymptotic equivalents; the general expression for  $A_n$  is exact. The Gaussian local limit is not asserted for fixed positive  $\varepsilon$ .

## References

- [1] Vladimir Reshetnikov, *ProveIt: Transseries project and series-and-transseries index*, repository documentation, inspected revision 0c973d8f5e7ef4550f4df1bf486d890037fd9f14, 29 September 2026. <https://github.com/VladimirReshetnikov/ProveIt/blob/0c973d8f5e7ef4550f4df1bf486d890037fd9f14/Analysis/Transseries/docs/series-and-transseries/README.md>. Inspection scope and status caveats are described in Appendix A.
- [2] *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, research package in ProveIt, 29 September 2026. Research Question 4, source lines 1467–1472, inspected at the revision above. [https://github.com/VladimirReshetnikov/ProveIt/blob/0c973d8f5e7ef4550f4df1bf486d890037fd9f14/Analysis/Transseries/docs/series-and-transseries/Critical\\_Hahn\\_Transseries\\_Beyond\\_Finite\\_Action\\_Folds/article.tex](https://github.com/VladimirReshetnikov/ProveIt/blob/0c973d8f5e7ef4550f4df1bf486d890037fd9f14/Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/article.tex).
- [3] *The Logarithmic Critical Endpoint: Convergent Lambert Charts, All-Order Sector Laws, and Sharp Action-Cutoff Corrections*, unpublished research draft prepared for Vladimir Reshetnikov, 29 September 2026. Inspected in the user’s Library under `article(20260929-201315).tex`. The scope statement explicitly treats the fixed endpoint and excludes the joint exponent/order limit. Used for scope comparison, not as a proof dependency.
- [4] *Marginal Critical Transseries: Convergent Lambert–W Charts, Universal Cutoff Corrections, and Certified Action Budgets*, unpublished research draft prepared for Vladimir Reshetnikov, 29 September 2026. Inspected in the user’s Library under `Marginal_Critical_Transseries.tex`. Its conclusion identifies uniform endpoint crossover as a further task. Used for scope comparison only.
- [5] NIST Digital Library of Mathematical Functions, Chapter 25, especially §25.12(ii), Eq. 25.12.12, and the zeta functional equation in §25.4. <https://dlmf.nist.gov/25.12.E12>. Accessed 29 September 2026. The standard polylogarithm expansion is the external special-function identity used in Proposition 3.2.
- [6] Svante Janson, Simply generated trees, conditioned Galton–Watson trees, random allocations and condensation, *Probability Surveys* **9** (2012), 103–252. DOI: 10.1214/11-PS188. <https://arxiv.org/abs/1112.0510>. Example 18.29 is a classical fixed cubic-tail antecedent of Gaussian/extreme-value separation. The moving-exponent local-limit proof in this article is given independently.